

AC-DC Converter Systems

Precision Power Conversion for Industrial, Defense & Mission-Critical Applications

**Reliable Power Conversion.
Stable High-Power DC Output.
Built for Advanced Applications.**



CRYONANO DC-DC Converter Systems deliver precise step-up, step-down, and isolated voltage conversion for sensitive electronic systems. Engineered for distributed power architectures, these converters provide stable, low-noise DC power across industrial, defense, aerospace, transportation, and embedded environments.

The portfolio spans from compact embedded modules to high-power scalable platforms, ensuring reliable performance in conditions where standard power systems cannot operate.

Notes

- Performance depends on load conditions, thermal design, and system integration
- Wide input models available for unstable or variable DC sources
- Custom compliance and qualification available for defense and industrial standards

KEY FEATURES

• High-Efficiency Power Conversion

Up to 96% efficiency with low output ripple and noise enables optimized high-density, low-loss operation.

• Flexible Input Architecture

Supports wide DC input ranges (12/24/28/48/72/110 VDC) with selectable models designed for distributed DC bus integration.

• Voltage Conversion Capability

Supports step-up, step-down, and galvanically isolated outputs with stable regulation for multi-rail and distributed system architectures.

• Integrated Protection System

Provides over-voltage, over-current, short-circuit, and thermal protection with active fault monitoring and shutdown.

• Compact & Scalable Design

Compact high-density form factors support scalable deployment from low-power modules to high-power platforms for embedded and system-level applications.

APPLICATIONS

- Industrial automation and control systems
- Medical and scientific instrumentation
- Railway and transportation electronics
- Embedded and distributed power systems
- Telecommunications infrastructure
- Energy storage and battery management systems (BMS)
- Defense and aerospace platforms



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SPECIFICATIONS

ELECTRICAL PERFORMANCE

Input Voltage Range	12–110 V DC (model dependent)
Input Range (Wide Models)	Up to 6:1
Output Voltage	Configurable (step-up / step-down / isolated)
Efficiency	Up to 96%
Power Range	300 W to 30 kW

OUTPUT CHARACTERISTICS

Output Type	Isolated / non-isolated
Output Ripple	Low-noise, low ripple
Regulation	High stability
Multi-Output	Supported (configuration dependent)

PROTECTION & MONITORING

Over-Voltage Protection	Integrated
Over-Current Protection	Integrated
Short-Circuit Protection	Integrated
Thermal Protection	Integrated
Fault Handling	Automatic shutdown & monitoring

MECHANICAL & THERMAL DESIGN

Form Factor	Compact module to rack/frame systems
Cooling	Convection / forced air (configuration dependent)
Power Density	High-density architecture
Build Type	Industrial / ruggedized

CONFIGURABILITY

Architecture	Options COTS / Modified / Fully Custom
Integration	Embedded / distributed systems
Output Configuration	Single / multi-rail
Customization	Voltage, cooling, form factor

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